AGENTVERSE 2026 · ROBOTICS

Rig Rust Repair

A robot dog that finds pipe corrosion on an offshore rig, fixes it, proves it's fixed, and asks a human when it isn't sure.



THE PROBLEM

Offshore pipework never stops rusting

Dangerous

Rope access and scaffolding, 30 m up, in salt spray, next to live hydrocarbons.

Costly

Every repair needs people flown offshore to do it by hand with remover, a wire brush and paint.

Never finished

Kilometres of pipe per platform, corroding again.

User: **asset-integrity and maintenance crews.** The worst thing a robot could do is **paint over rust and hide it.**

WHAT WE BUILT

Go2 quadruped + 0–0.45 m lift mast + UR3 arm

On a full offshore rig in Isaac Sim 6.2, rendered on an NVIDIA GB10 and streamed live to the browser.

PATROL → DETECT → SPRAY → SOAK 30 s → SCRUB · PPO →
VERIFY · detector + Gemini → PAINT → DONE

at any time: RE-PLAN BACK_OFF SECOND_VIEW HOLD PAUSE_PERSON
ESCALATE_TO_HUMAN

Every arrow is triggered by what the camera sees or what the brush feels, and every transition is logged with its evidence.

DEMO · ONE TAKE · ≤ 60 S

▶ Play video

Watch the HUD: detector mask → target grid · an unscripted new rust patch → RE-PLAN · BACK_OFF / HOLD / PAUSE

File: **TBD** (~/Downloads/rig-videos/) · fallback: rig.zeecloud.org live stream

PERCEPTION-DRIVEN ACTION

Pixels become the robot's work plan

Camera frame → 96 px patches → Heatmap → Mask → Ray–cylinder projection → (θ, z) grid on the pipe → Spray path · scrub targets · done check

MobileNetV4-S + LoRA

0.26 M trainable parameters · 1.1 ms per crop on CPU · recall-first threshold with an “unsure” band → second view

0.977

F1 on 881 real photos (recall 0.983, AUROC 0.998)

13/13

held-out Isaac renders (10/13 before sim-style data generated from the scene's own textures)

SCRUB POLICY · PPO IN ISAAC LAB, 2,048 ENVS

v1 cleaned everything by pressing too hard

RIG GEOMETRY, NEW RUST PATCH EACH EPISODE	V1	V2 (FORCE-LIMITED)
Success (< 5 % rust left)	100 %	97.6 %
Episodes over the 40 N limit	**76.8 %**	**1.0 %**
Mean / p95 episode-max force	**49.9 / 70.1 N**	**29.9 / 35.7 N**
Mean time to clean	5.2 s	6.8 s

Wear model: removal ∝ force × speed, 4× faster after a 30 s soak. v2 adds a quadratic over-force penalty and a 38 N setpoint cap. A slightly slower scrub that doesn't damage the pipe is the better trade.

CRITICAL FAILURE MODES

When in doubt: look again, hold, or ask a human

Paints over rust	VERIFY: < 5 % rust, ≥ 80 % visible · Gemini veto · 3 failed cycles → **ESCALATE_TO_HUMAN** with evidence
Force > 40 N	**BACK_OFF**: retract 5 cm, press × 0.7 · 3× → escalate
Ambiguous (p 0.35–0.65)	up to 2 **SECOND_VIEW**s → Gemini → human
Camera blocked	**HOLD** · escalate after 5 s
Person < 3 m	**PAUSE_PERSON** · resume when clear
New rust mid-run	**RE-PLAN**: spray it, extend the soak

Every response is on the HUD and in decisions.jsonl, with the confidence and force behind it.

EDGE AUTONOMY

The fast loop never leaves the robot

On the robot · CPU

- Detector: 1.1 ms
- Scrub policy: 0.012 ms*
- Orchestrator: 10 Hz

Cloud · optional

- Gemini Robotics-ER second opinion
- Asynchronous; latency **TBD**
- Offline → same behaviour as without it

Training only

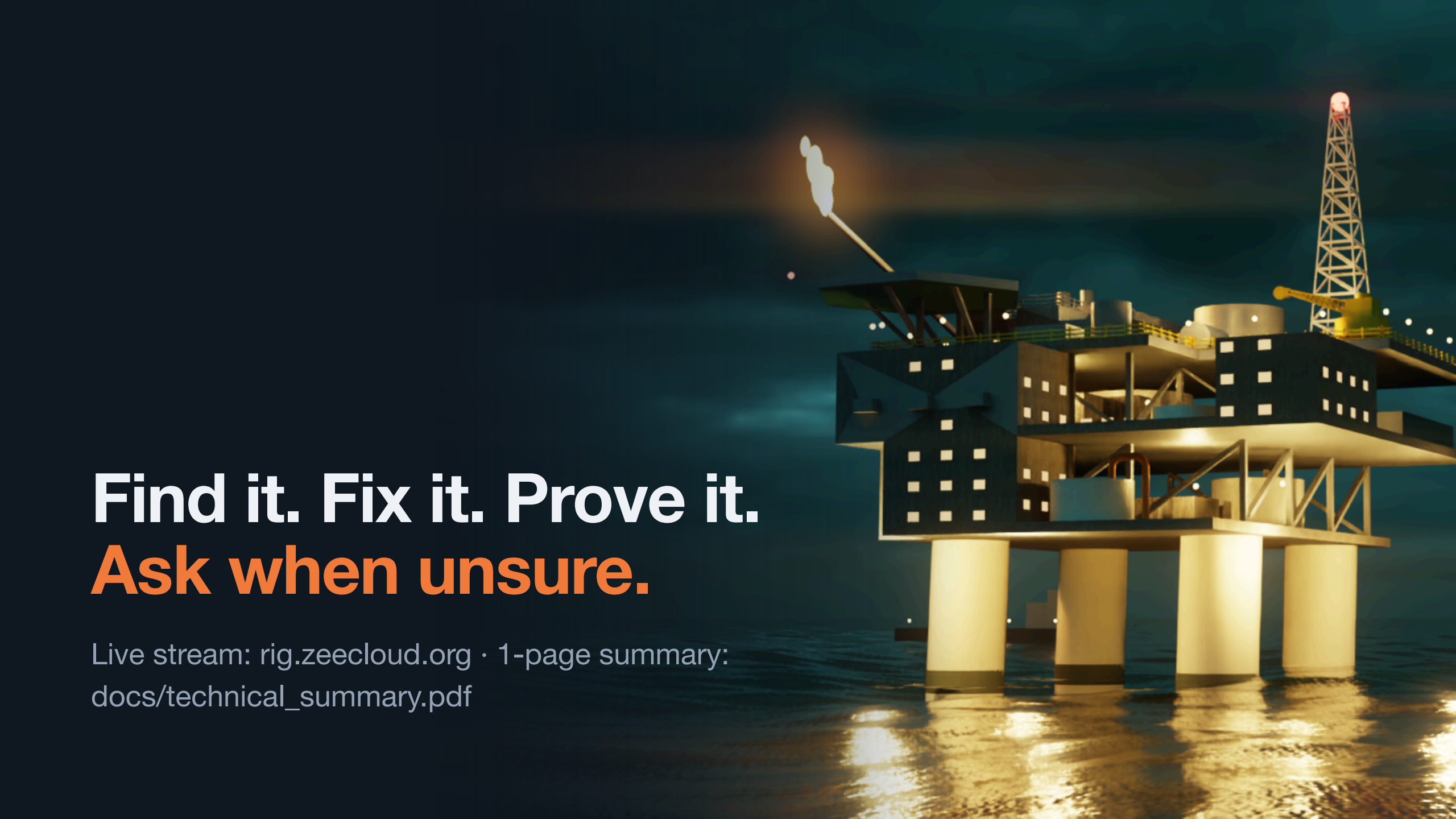
- Modal A10G: detector
- Modal L40S ~45k steps/s
- GB10 ~15k steps/s (and the RTX render + stream)

*ONNX, batch 1, measured on a laptop CPU.

LIMITS, HONESTLY

What this is, and what comes next

- Simulation only. The demo bridge poses the robot kinematically, and the brush force is a spring model.
- The scrub policy trains on grid state; in deployment the localizer fills it.
- The sim-render test set is small (13 crops), and there are no real painted-steel negatives.
- Next: legs that walk under arm load, a real force-torque sensor, real painted-steel data.



Find it. Fix it. Prove it.
Ask when unsure.

Live stream: rig.zeecloud.org · 1-page summary: docs/technical_summary.pdf